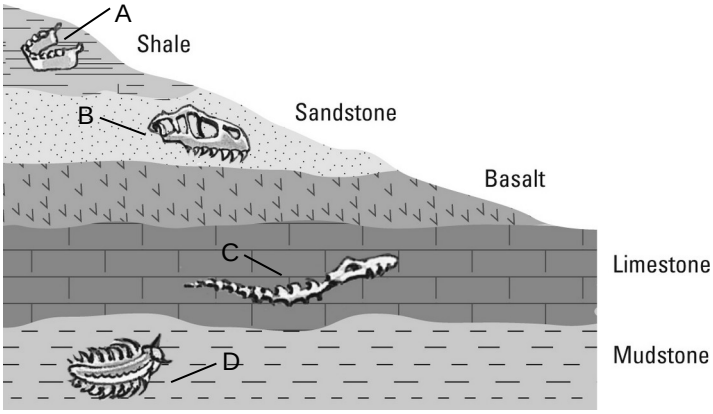
Ages of fossils

Answers

Relative ages

1. The following cliff face diagram contains numerous rock layers and fossils.



(a) Identify the youngest and oldest sedimentary rock layer in this sequence.
Youngest = shale; oldest = mudstone

(b) Using the code letters (A–D), arrange the fossils in order from oldest to youngest.
D, C, B, A

(c) Explain why there are no fossils in the basalt rock layer.
Basalt is an igneous rock that was once molten. Fossils are usually destroyed by molten rock.

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Radiometric C-14 dating

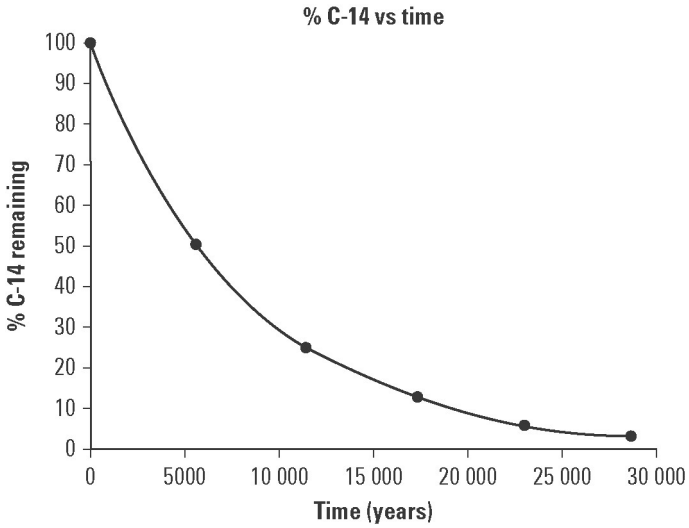
2. The absolute ages of recent carbonaceous fossils can be determined using radiocarbon dating.

The half-life of carbon-14 (a radioactive isotope) is 5730 years. The following table shows how the percentage of C-14 present in a fossil decreases over time.

Time (years)	0	5730	11 460	17 190	22 920	28 650
% C-14 remaining	100	50	25	12.5	6.25	3.125

(a) Complete the table.

(b) Plot a line graph of this data on the grid below.



(c) The C-14 content of a fossil was found to have decreased by 36% since the organism died. Determine the approximate age of the fossil.

64% carbon-14 remains

From graph: Age = 4000 years

(d) A fossil is approximately 10 000 years old. What percentage of C-14 still remains?

30% C-14 remains (from graph)

(e) Explain why it is difficult to date fossils that are more than 60 000 years old using radiocarbon dating.

After 60 000 years very little C-14 remains to measure.

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